

CUSTOMER RETENTION ANALYSIS

Reducing Customer Churn: A Data-Driven Retention Strategy

Predicting churn risk, segmenting the customer base, and prioritizing retention spend for maximum ROI

By Rhythanyaa Nagaraja | Telco Customer Dataset | Python, pandas, scikit-learn

Turning Raw Records Into a Modeling-Ready Dataset

THE DATASET

7,043

customers

21

features per customer

Source: IBM Telco Customer Churn dataset — account details, billing, subscribed services, and churn outcome.

1

Remove Duplicates

Checked for and dropped exact duplicate customer records.

2

Fix TotalCharges column

Converted from text to numeric; blank entries (zero-tenure customers) handled as missing.

3

Drop Missing Rows

Removed the small number of rows with missing values after cleanup.

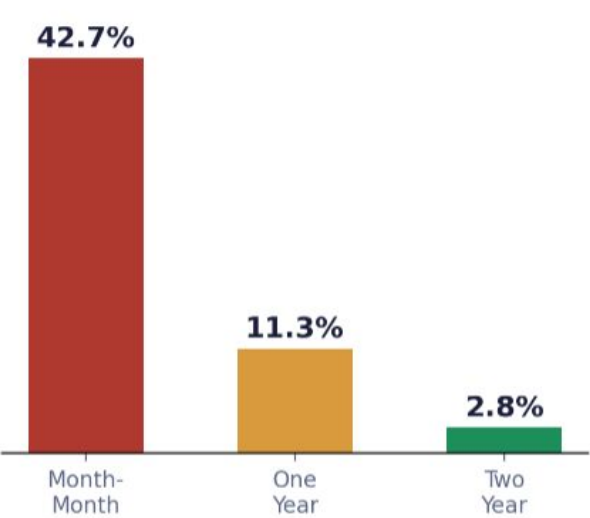
4

Encode Categories

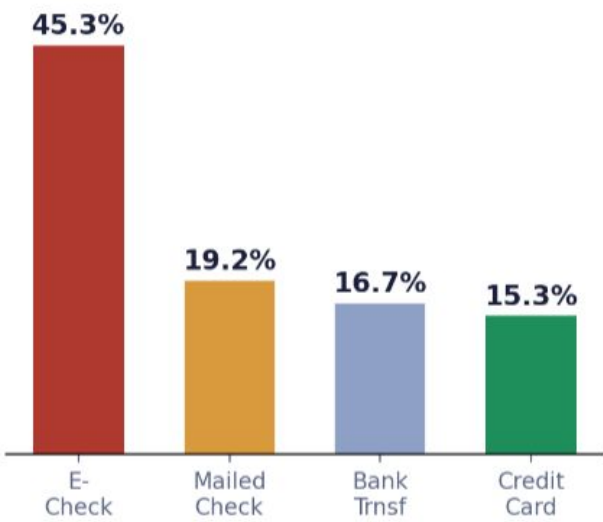
One-hot encoded contract type, payment method, and service add-ons for modeling (0 = No, 1= Yes)

Customer Behavior Trends

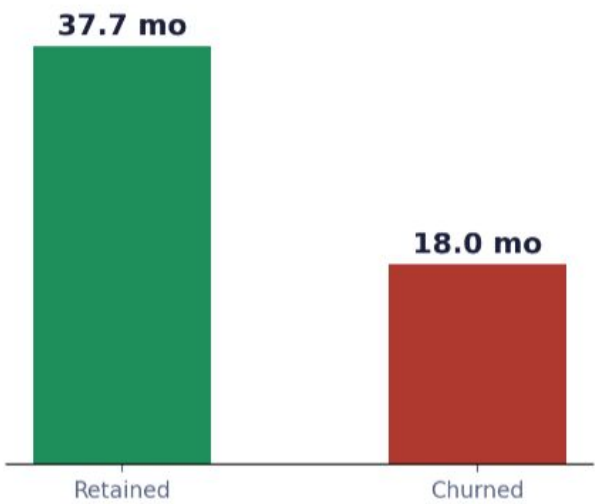
Churn by Contract Type



Churn by Payment Method



Tenure: Churned vs. Retained



Takeaway: churn is concentrated in month-to-month, electronic-check, early-tenure customers, the same customer profile shows up across all three signals, which pointed directly at what the model needed to catch.

THE PROBLEM

As a consultant: identify the key drivers of churn, quantify the revenue at risk, and recommend a targeted retention strategy that maximizes ROI given a limited budget.

26.6%

of customers churn overall, roughly 1 in 4. With a limited retention budget, a company cannot afford to treat every customer the same way; it needs to know exactly who to prioritize.

Predicting Which Customers Are at Risk

A Decision Tree model was trained to predict each customer's probability of churning. It works by learning simple, human-readable rules from the training data, splitting customers into groups based on the factors most linked to churn, like contract type, tenure, and monthly charges, until it can assign each customer a risk score.

Top predictor: contract type, month-to-month customers are far riskier than those on longer contracts. Short tenure (under 6 months) is the next strongest signal

Why a Decision Tree?

Unlike a lot of machine learning models, a decision tree's logic can be traced step by step, which makes it possible to explain exactly why a given customer was flagged as high-risk. That transparency matters for a business tool that needs buy-in from people who aren't data scientists.

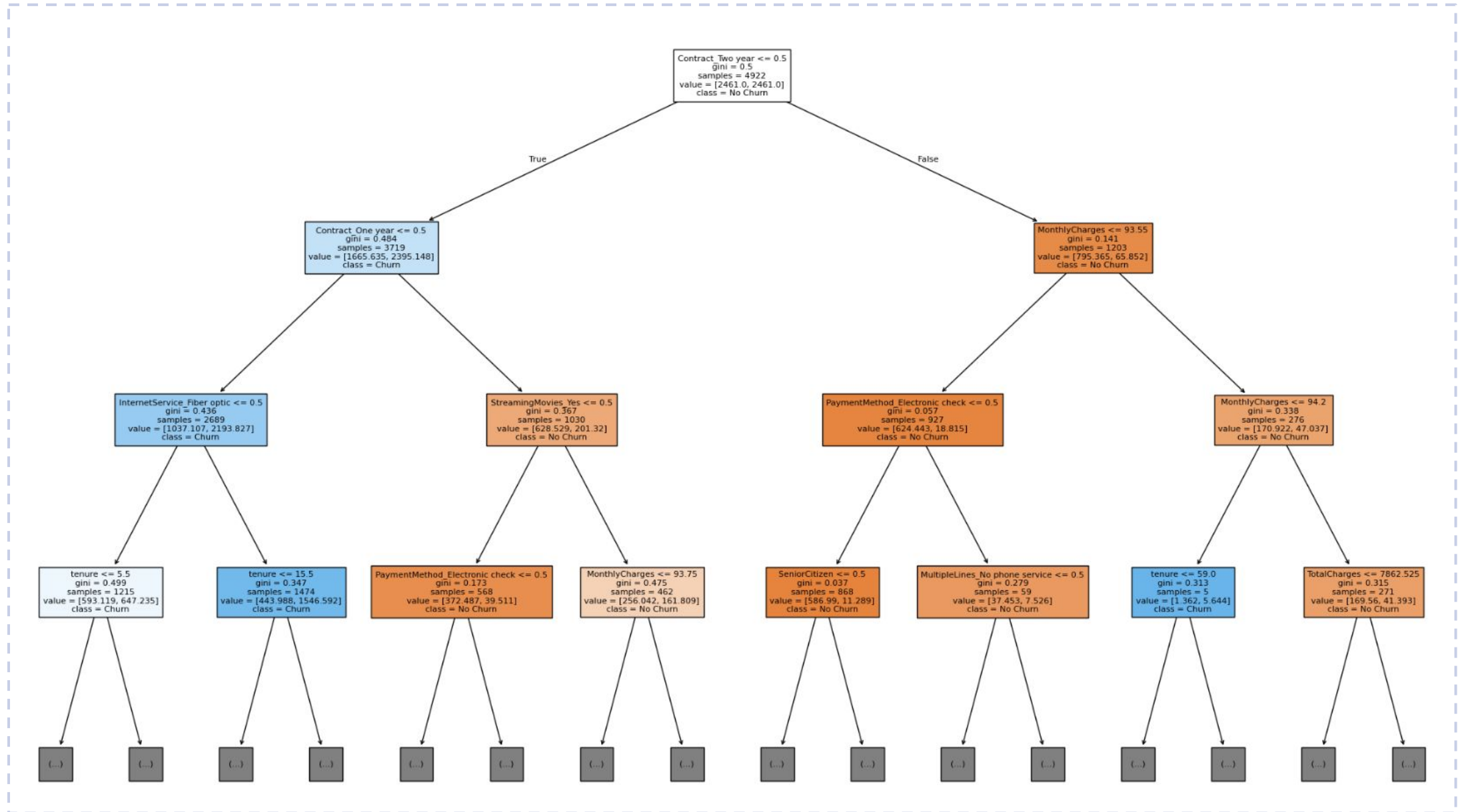
KEY RESULT

78%

of customers who actually churned were correctly flagged as high-risk by the model

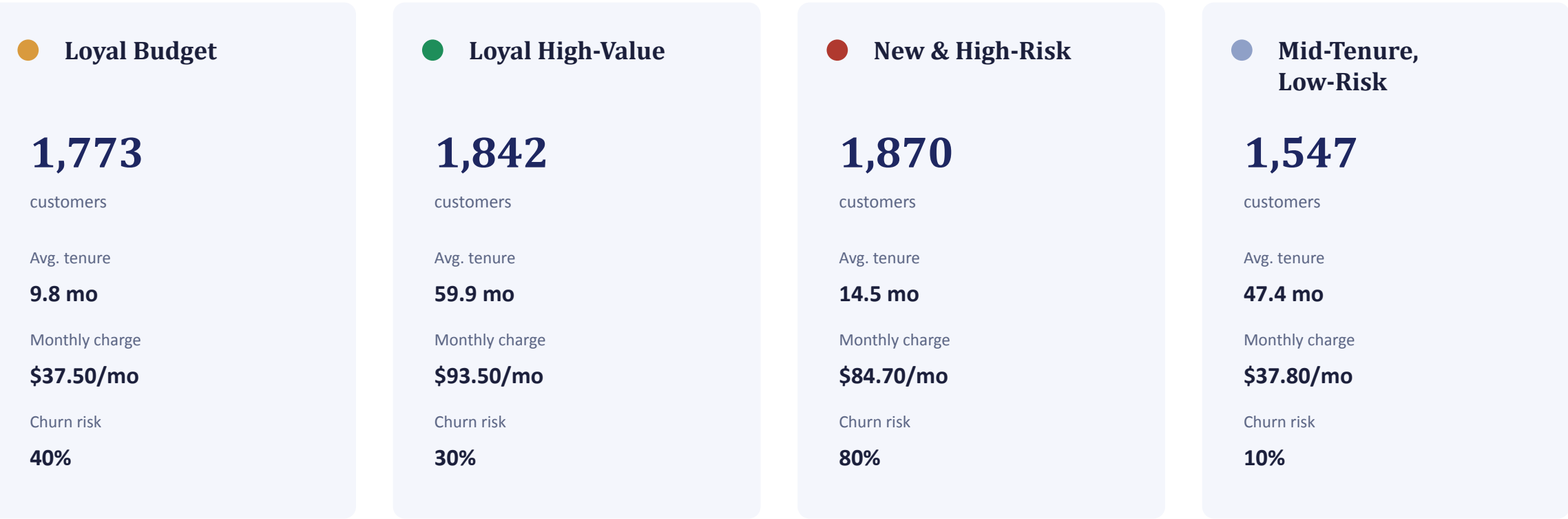
Why this matters: missing an at-risk customer costs their entire future revenue. Catching most of them, even if a few flags turn out to be false alarms, protects far more revenue than it costs.

How the Decision Tree Makes a Call



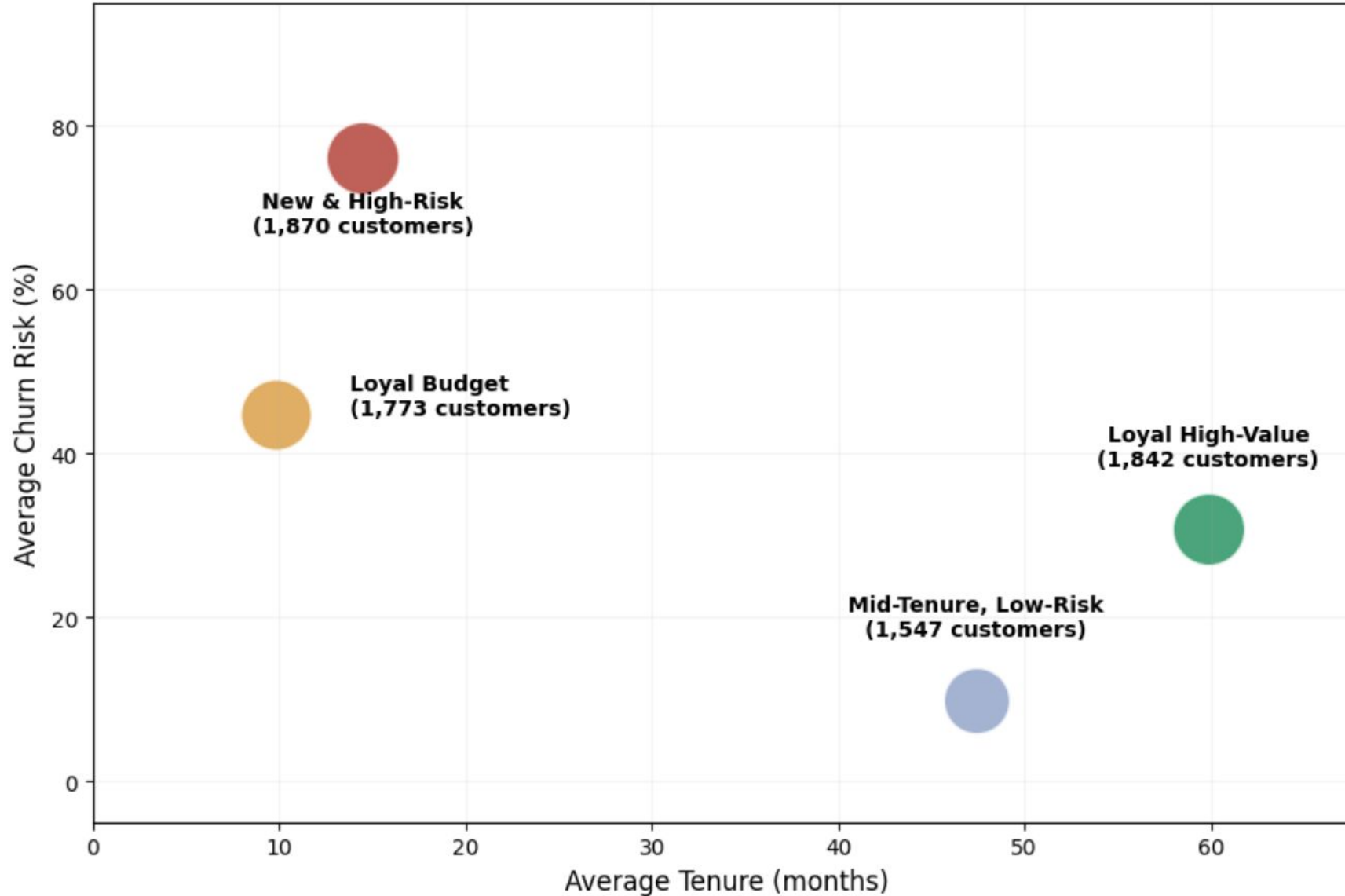
Grouping Customers by Value and Risk

K-Means clustering grouped all customers into four segments based on tenure, monthly spend, total spend, and predicted churn risk, so each segment reflects a distinct customer profile. This would help us create targeted plans to retain our customers.



Visualizing the Four Segments

Customer Segments: Tenure vs. Churn Risk



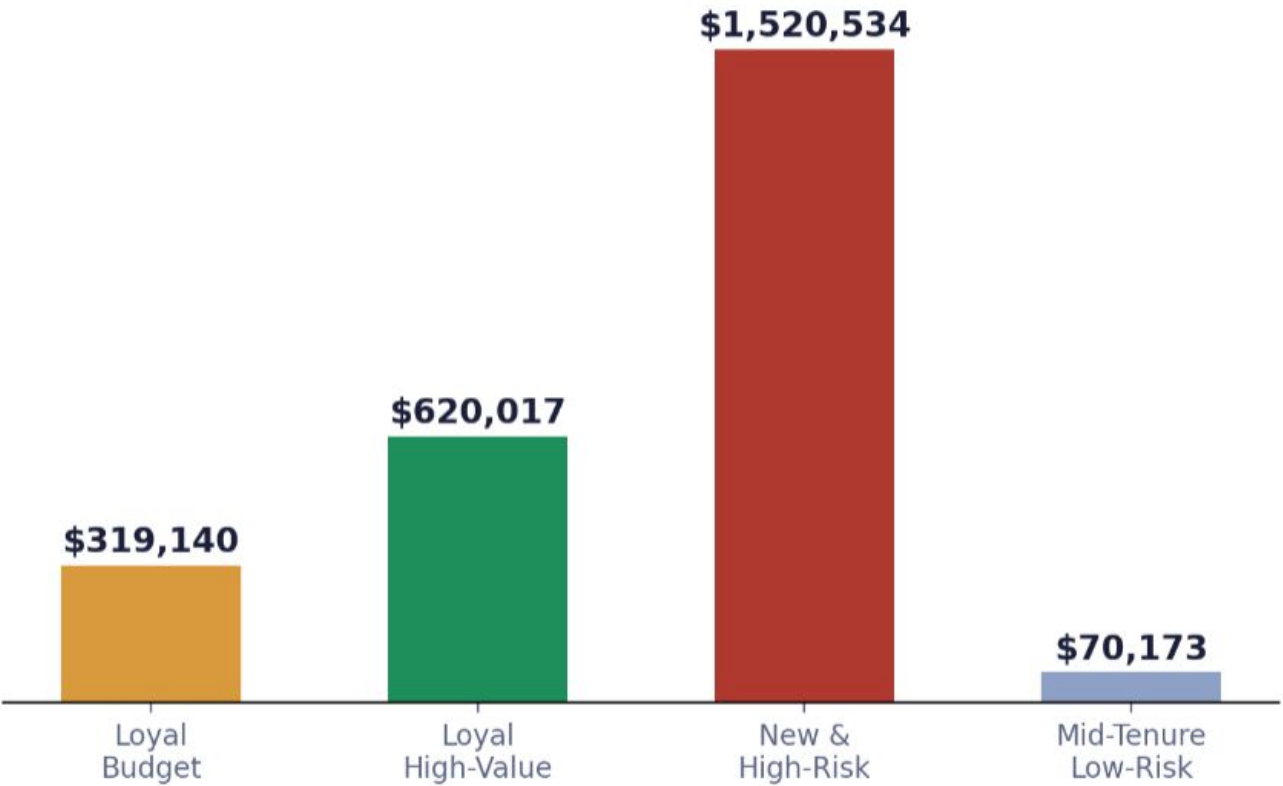
What Happens If Nothing Changes

PROJECTED ANNUAL

\$2.53M

in expected revenue at risk across the customer base if no retention action is taken

Calculated as each segment's annual customer value weighted by its average churn probability, the revenue a company should expect to lose without intervening.



Expected Value at Risk = (Monthly Charge × 12) × Avg. Churn Risk × Number of Customers

Segment	Customers	Monthly Charge	Annual Value / Customer	Avg. Churn Risk	Expected Value at Risk
Loyal Budget	1,773	\$37.50	\$450.00	40%	\$319,140
Loyal High-Value	1,842	\$93.50	\$1,122.00	30%	\$620,017
New & High-Risk	1,870	\$84.70	\$1,016.40	80%	\$1,520,534
Mid-Tenure, Low-Risk	1,547	\$37.80	\$453.60	10%	\$70,173
Total	7,032				\$2,529,864

Worked example (New & High-Risk): $\$84.70 \times 12 = \$1,016.40$ annual value $\rightarrow \times 80\%$ risk = $\$813.12$ at risk per customer $\rightarrow \times 1,870$ customers = $\$1,520,534$

What to Do, and What It's Worth

1

Target New & High-Risk first

Highest ROI by far: \$140K spend → \$1.38M net savings.

2

Light-touch Loyal High-Value

Fewer at risk, but worth more each; loyalty perks protect \$482K.

3

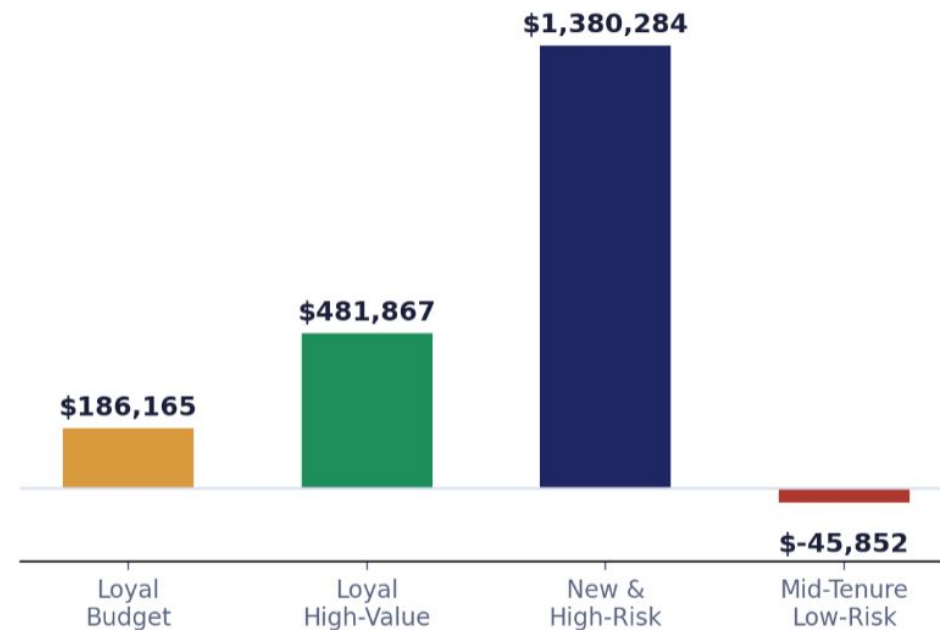
Low-cost outreach to Loyal Budget

Moderate risk, lower value; still nets \$186K.

4

Skip Mid-Tenure, Low-Risk

Only 10% risk, spending here loses money (\$46K net).



Bottom line: targeting the three ROI-positive segments delivers an estimated \$2.05M in net savings on ~\$387K in retention spend, roughly a 5.3x return.

THE FIX

The Plan for New & High-Risk Customers

This is the top-priority segment: 1,870 customers, new to the company, already spending well, and at 80% average churn risk.

1

Incentivize a 1-Year Contract Upgrade

Contract type is the single strongest churn driver: month-to-month churns at 42.7% vs. 2.8% on two-year.

Offer a modest discount (~10%) tied specifically to switching off month-to-month onto a 1-year term.

2

Push Autopay Enrollment

Electronic check payers churn at 45.3%, nearly 3x every other payment method, the second-strongest signal found.

A small one-time credit for enrolling in autopay, low cost, removes a proven churn correlate.

3

Structured 90-Day Onboarding

This segment's defining trait is short tenure; new customers haven't built loyalty yet and risk is highest early on.

Scheduled check-ins at day 7, 30, and 60 to catch issues before they become a reason to leave.

4

Bundle a Free Trial of Tech Support

Customers with active support services showed lower churn in the exploratory analysis of this dataset.

Offer a no-cost trial period of tech support to increase engagement and switching cost.

Impact: \$140K in retention spend on this segment is projected to save \$1.52M, a net gain of \$1.38M, roughly a 9.8x return.

CONCLUSION

Churn Is Predictable, and That Makes It Actionable

With a \$75-per-customer retention offer targeted at the right segments, this strategy is projected to save \$2.05M annually, a 5.3x return on retention spend.

Thank you.